



Department of Education

**STEM Education**  
**National Schools of Excellence**  
**STEM Biology Syllabus**  
**Grade 11**  
**Term 2**

**DOMAIN**      **Phylum**

**SCIENCE**      **Biology**

**Technology**

**Engineering**

**Mathematics**

Issued free to schools by the Department of Education

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## **Acknowledgements**

This STEM Biology Syllabus for Grade 11 constitutes ten (10) Thematic Areas (Thema) and is intended for students enrolling for STEM Education. In this booklet only three Thema are presented.

Ultima Thulë Ltd was engaged by the Department of Education to develop the STEM Curriculum. The content was generated by writers from universities, secondary schools, new university graduates with knowledge in the subject matter and industry subject experts. Quality assurance groups and the Science Subject Advisory Committee have also contributed to the development of this syllabus.

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## SECRETARY'S MESSAGE

This Biology Syllabus is to be used exclusively by teachers of Biology to teach Grade 11 Students enrolling for the STEM Education through the Schools of Excellence throughout Papua New Guinea. The STEM Biology Bridging Syllabus ensured that Grade 11 students enrolling for STEM Education were grounded in the basic concepts espoused in Biology before undertaking the advanced concepts introduced in this syllabus.

The syllabus conforms to the visions of the National Education Plan and the School of Excellence Policy. Furthermore, it is undergirded by a policy and legislative matrix that were issued by the National Government over the last decades, such as the Vision 2050 and Development Strategic Plan 2030. The Preamble of the National Constitution, especially the First Directive of the *National Goals and Directive Principles* makes a clarion call for the Integral Human Development, which the Department took heed of to ensure, through this new syllabus, that *all forms of beneficial creativity, including sciences and cultures, [are] to be actively encouraged*. A principle object of the *National Education Act 1983 (amended 1995)* reiterates this virtue of Integral Human Development (Section 4).

In accordance with the widespread demand for a home-grown curriculum to replace the existing one with the view to position the country as a bastion of innovation in the new century the Secretary of Education has embarked on introducing the STEM Curriculum to the Schools of Excellence. It is hoped that students coming through the STEM Education will not only gain productive employment in the knowledge-based economy but become creators, innovators and inventors of knowledge and ultimately commercialize their creativity through development of their intellectual property rights.

In addition, there are four core elements espoused in this syllabus, insofar as student development is concerned: Students are expected to gain knowledge and skills and apply them to solve every day problems whilst appreciating Biology as a fundamental field of science.

This STEM Biology Syllabus for Grade 11 constitutes ten (10) Thematic Area (Thema) and is intended for students enrolling for STEM Education. In this booklet only three Thema are presented. By studying these Thema students are expected to gain the foundational knowledge needed in Biology.

The success of this new Curriculum is highly dependent on the Teachers. They play a pivotal role by being innovative and keeping abreast of new information on scientific research and innovations. The challenge for teachers of Biology is to engage students to learn realistic context for increased and better understanding and appreciate the relevance of Biology to human lives, including good health and environment.

In order to ensure Teachers discharge their responsibilities adequately this syllabus is accompanied by a Teacher's Guide and Teacher's Resource Book which provides sufficient teaching and assessment materials to help Teachers.

I commend and approve this syllabus as the official curriculum for Biology to be used for the Grade 11 STEM Education in the Schools of Excellence throughout Papua New Guinea.

DR UKE KOMBRA, PHD  
Secretary for Education

# INTRODUCTION

Biology is the study of living organisms and how they interact at multiple levels to sustain life. The basic building block of life is considered to be the cell. The hereditary units are the genetic sequences and these are encased within the cells. This genetic information ensures the hereditary traits necessary for survival of the species is passed on from parents to progenies. Biology is divided into different fields of study. For instance, botany is the study of plants whilst zoology involves the study of animals. Micro-organisms are studied in microbiology. Recent fields that emerged in the last few decades include molecular biology, genomics, and so on. Since Biology is an experimental science, laboratory work becomes an essential part of the syllabus. Practical work can be used to enable students to investigate the properties and reactions of substances and to provide opportunities for students to learn and test the principles and concepts of Biology. Since Biology is a highly specialized subject students undertaking Biology are expected to possess high levels of cognitive and numeracy competency, and a firm grasp of the language skills. The modules in this Syllabus are sequenced to guide teachers of Biology through the process of imparting basic Biology knowledge and skills to students.

The Grade 11 Biology Syllabus entails ten (10) Thematic Areas (Thema) and these are listed below. Each Thema has a single Module and three topics. Over time different Modules will be introduced.

| THEMA               | MODULE                                               | TOPICS                                                                      |
|---------------------|------------------------------------------------------|-----------------------------------------------------------------------------|
| BIODIVERSITY        | Agricultural Biodiversity                            | 1. Agricultural Biodiversity in Papua New Guinea<br>2. Genetic Diversity    |
|                     | Ornithology                                          | 1. Birds of Paradise<br>2. Inspirations from bird behaviour                 |
| BIOTECHNOLOGY       | Concepts and Applications of Biotechnology           | 1. Recombinant DNA Technology<br>2. Industrial Biotechnology                |
| HEALTH AND MEDICINE | Medicine and Health Sciences                         | 1. Pharmacology<br>2. Medical Diagnostic Investigations<br>3. Public Health |
| EARTH SCIENCE       | Environment                                          |                                                                             |
| WASTE MANAGEMENT    | Biological Wastes                                    |                                                                             |
| INFRASTRUCTURE      | Natural Inspirations in Infrastructure               |                                                                             |
|                     | Agricultural Infrastructures                         |                                                                             |
| ENERGY              | Cell Energy and Power                                | 1. Photosynthesis<br>2. Catabolism                                          |
| BATTERY TECHNOLOGY  | Battery Technology Application in Biological Systems |                                                                             |

|                         |                                                |                                                                                                                                                                                                                           |
|-------------------------|------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| INFORMATION SYSTEM      |                                                |                                                                                                                                                                                                                           |
| COMPUTER SCIENCE        |                                                |                                                                                                                                                                                                                           |
| ARTIFICIAL INTELLIGENCE | Biological Concepts in Artificial Intelligence | <ol style="list-style-type: none"> <li>1. Neurobiology</li> <li>2. Neurobiological models in Artificial Intelligence</li> </ol>                                                                                           |
| ENTERPRENEURSHIP        |                                                | <ol style="list-style-type: none"> <li>1. Intellectual Property (Patents, Copyrights, Trademarks, etc)</li> <li>2. How to protect Intellectual Property</li> <li>3. How to commercialize Intellectual Property</li> </ol> |

Since Biology is a highly specialized subject students undertaking Biology are expected to possess high levels of cognitive and numeracy competency, and a firm grasp of the language skills.

The modules in this Syllabus are sequenced to guide teachers of Biology through the process of imparting basic Biology knowledge and skills to students.

## RATIONALE

There is a confluence of events that necessitated the need to introduce STEM Curriculum to the Schools of Excellence:

1. The School of Excellence Policy that was launched by Prime Minister Honourable James Marape, MP, in June 2020, calls for the establishment of the Schools of Excellence and the introduction of STEM Curriculum;
2. The Government has decided to replace the Observation Based Education to Standards Based Education in 2019 and tasked the Department of Education to introduce the STEM Curriculum as part of its reform agenda;
3. There is public outcry, both from public and private sectors, over the quality of graduates coming through the National Education System; and
4. The changing landscape in the technology domain globally and the pressing need to produce the next generation of workforce that is agile and can adopt to the changing circumstances as well as become enterprising individuals through their creation of intellectual properties.

The combination of these factors has led the Department of Education to adopt the STEM Curriculum and to introduce this to the Schools of Excellence as a necessary disruptive intervention in an effort to develop the next generation of national intellectual capital and to drive the growth and prosperity of the country through creation of intellectual properties.

In developing the STEM Curriculum, it was realized that the Grades 9 and 10 Science Syllabi does not provide a firm foundation to launch students into STEM Curriculum at the Schools of Excellence. Consequently, a STEM Bridging Program was necessary. The STEM Bridging Syllabus is a bridge across the divide that Grade 11 Students enrolling for STEM Education must cross.

In the Biology Bridging Syllabus students were introduced to the basic concepts and knowledge of Biology in order for them to develop an appreciation of the discipline in a broader context. In this syllabus students are introduced to advanced concepts in Biology as they are applied to the ten Thema.

The introduction of the STEM Curriculum, especially the Bridging Program, will cause consequential changes to the Upper Classes of STEM Education and the Lower Secondary Syllabi. These syllabi will be introduced in the course of time.

## NEW NOMENCLATURE

As part of the overall Curriculum Development new nomenclature were introduced. These are explained below:

| Term   | Explanation                                                                                                                                           |
|--------|-------------------------------------------------------------------------------------------------------------------------------------------------------|
| Domain | Each STEM area is defined as <i>domain</i> (e.g. Science Domain, Technology Domain, etc.)                                                             |
| Phylum | What would normally be regarded as “subject” (e.g. Biology, Physics, Chemistry, etc.) is now regarded as <i>phylum</i> ; the plural is <i>phyla</i> . |
| Thema  | Thematic areas of studies                                                                                                                             |
| Module | A specific concept within the Thema which is being studied                                                                                            |
| Topic  | A specific aspect of the Module that is being studied                                                                                                 |



## **STEM Domains**

At this juncture of our national development there is no other task before us more important than deploying a robust STEM Curriculum that develops the next generation of highly educated students to enter the work force. By building on the best of current practices and standards the STEM Curriculum aims to take us, beyond the constraints of the existing structures of schooling, towards a shared vision of excellence nationally and marketing the nation globally in terms of Science, Technology, Engineering and Mathematics. In this STEM Curriculum, there are ten (10) Thema (Thematic Areas) introduced.

### **Science**

It is the study of the natural world, including the laws of the nature associated with physics, chemistry and biology and the treatment or application of facts, principles, concepts or conventions associated with these disciplines. Science is both a body of knowledge that has been accumulated over time and a process-scientific inquiry-that generates new knowledge. Knowing from science informs the engineering design process. In this STEM Curriculum the Science Domain has three Phyla – Biology, Chemistry and Physics.

### **Technology**

Comprises the entire system of people and organizations, knowledge, processes, and devices that go into creating and operating technological artifacts, as well as the artifacts themselves. Throughout history humans have created technology to satisfy their wants and needs. Much of modern technology is the product of science and engineering and technological tools used in both fields. There are two Phyla – Information Technology and Computer Sciences – introduced in this STEM Curriculum for the Technology Domain.

### **Engineering**

Concerns both the knowledge about the design and creation of products and a process for solving problems. The design process must result in a final product that must operate or function under various constraints. One constraint in engineering design is the laws of nature. Engineers must learn to harness the natural forces at work around them and through their design process develop products that must operate in harmony with nature. Other constraints include things such as time, money, available materials, ergonomics, environmental regulations and manufacturability. Engineering utilizes concepts in science and mathematics as well as technological tools. There are three (3) Phyla introduced in the Engineering Domain – Civil, Mechanical and Electrical.

### **Mathematics**

It is the study of patterns and relationships among quantities, numbers, and shapes. Specific branches of mathematics include arithmetic, geometry, algebra, trigonometry and calculus. Mathematics is used in science and in engineering. In this STEM Curriculum, Mathematics is considered as a Service Domain; specific topics tailored towards specific Thema are also considered relevant to undergird the relevance and applicability of Mathematics Domain to other fields of knowledge.

## STEM BIOLOGY CONTENT OVERVIEW – TERM 2

**Table 1.1: Thema Sequence and Duration**

| Grade 11 Term 2: STEM-proper Course     |                                                 |                                                                              |
|-----------------------------------------|-------------------------------------------------|------------------------------------------------------------------------------|
| <b>BIODIVERSITY</b><br>(3 Weeks)        | 11.1 Agricultural Biodiversity                  | 1. Agricultural Diversity in Papua New Guinea<br>2. Genetic Diversity        |
|                                         | 11.2. Ornithology                               | 1. Bird of Paradise<br>2. Inspirations from Bird Behaviour                   |
| <b>BIOTECHNOLOGY</b><br>(3 Weeks)       | 11.1 Concepts and Applications of Biotechnology | 1. Recombination DNA Technology<br>2. Industrial Biotechnology               |
| <b>HEALTH AND MEDICINE</b><br>(3 Weeks) | 11.1 Health and Medical Biology                 | 1. Pharmacology<br>2. Medical Diagnostics Investigations<br>3. Public Health |

**Table 1.2 Sequencing & Benchmarking STEM Chemistry Content Standards**

| Setting STEM Chemistry Content Standards for Progressive Learning & Assessment |                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                                                                                                                                                             |                                        |
|--------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------|
| Thema                                                                          | Biodiversity                                                                                                                                                                                                                                                                                                                                                                           | Broad Content Standard                                                                                                                                                                                                                                                                                                                                                                                      |                                        |
| <b>Module 11.1</b>                                                             | Agricultural Biodiversity                                                                                                                                                                                                                                                                                                                                                              | <b>11.1:</b> Recognize and appreciate the rich biodiversity within Papua New Guinea and how to industrialize our biodiversity cautiously and conservatively from outside influence.                                                                                                                                                                                                                         |                                        |
| Topics                                                                         | Performance Indicators<br>Students are expected to:                                                                                                                                                                                                                                                                                                                                    | Module Benchmark                                                                                                                                                                                                                                                                                                                                                                                            | Assessment Tasks                       |
| 1.Agricultural Diversity in Papua New Guinea                                   | <b>11.1.1a:</b> Describe the four levels of organisation of biodiversity.<br><b>11.1.1b:</b> Describe the diverse agricultural biodiversity of Papua New Guinea<br><b>11.1.1c:</b> Identify and describe the wild relatives of agricultural crops native to Papua New Guinea<br><b>11.1.1d:</b> Identify and describe the limitations to agricultural development in Papua New Guinea. | Students master the ability and knowledge to: <ul style="list-style-type: none"> <li>Identify and describe the arms of agricultural biodiversity that contribute genetic resources for food and agriculture as well as ecosystem support services.</li> <li>Evaluate and analyze the vast genetic diversity within Papua New Guinea and by what means to conserve it as well as to manipulate it</li> </ul> | <b>Research Project + Written Test</b> |

|                                     |                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                                                                                                                                                                                                        |                                                         |
|-------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------|
| 2. Genetic Diversity                | <p><b>11.1.2a:</b> Identify and describe the genetic diversity of native agricultural crops in Papua New Guinea compared to the world and how to conserve native species.</p> <p><b>11.1.2b:</b> Recognize the importance of genetic sequencing of native agricultural crops in PNG and their applications to address climate resilience and food security.</p>                                       | utilizing technology for purposes of food security and enhancing productivity.                                                                                                                                                                                                                                                                                         |                                                         |
| <b>Thema</b>                        | <b>Biodiversity</b>                                                                                                                                                                                                                                                                                                                                                                                   | <b>Broad Content Standard</b>                                                                                                                                                                                                                                                                                                                                          |                                                         |
| <b>Module 11.2</b>                  | <b>Ornithology</b>                                                                                                                                                                                                                                                                                                                                                                                    | <b>11.2:</b> Understand the significance of Birds of Paradise to Papua New Guineans and glean the importance of the behaviours of birds and their implications in geopolitics and organizational strategies                                                                                                                                                            |                                                         |
| <b>Topics</b>                       | <b>Performance Indicators</b><br>Students are expected to:                                                                                                                                                                                                                                                                                                                                            | <b>Module Benchmark</b>                                                                                                                                                                                                                                                                                                                                                | <b>Assessment Tasks</b>                                 |
| 1. Birds of Paradise                | <b>11.2.1a:</b> Analyze the importance of the Ornithology of the Birds of Paradise as our National Emblem, the species diversity and their natural character or behaviour.                                                                                                                                                                                                                            | Students' ability to demonstrate knowledge and skills in; <ul style="list-style-type: none"> <li>Describe the ornithology of the Birds of Paradise species and their significance in Papua New Guinea.</li> <li>Discuss and analyze the geopolitical significance of inspirations from bird behaviors that are unique to human or organizational strategies</li> </ul> | <b>Research Project</b><br><br>+<br><b>Written Test</b> |
| 2. Inspirations from Bird Behaviour | <p><b>11.2.2a:</b> Analyze the nesting and predatory behaviour of birds that inspire geopolitical strategies utilized by organizations.</p> <p><b>11.2.2b:</b> Appreciate the inspirations from bird behaviour that has significant applications in diverse settings.</p> <p><b>11.2.2c:</b> Give examples: Mendondi and Tenduage Strategy as applied in Organizational and Corporate Strategies.</p> |                                                                                                                                                                                                                                                                                                                                                                        |                                                         |

| Thema                              | Biotechnology                                                                                                                                                                                                                                                                              | Broad Content Standard                                                                                                                                                                                                                                                                                                                                 |                     |
|------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|
| <b>Module 11.1</b>                 | Concepts and Applications of Biotechnology                                                                                                                                                                                                                                                 | <b>11.1:</b> Appreciate the concepts and applications of biotechnology in the medical or pharmaceutical and agricultural industries.                                                                                                                                                                                                                   |                     |
| Topics                             | Performance Indicators<br>Students are expected to:                                                                                                                                                                                                                                        | Module Benchmark                                                                                                                                                                                                                                                                                                                                       | Assessment Tasks    |
| 1. Recombination of DNA Technology | <b>11.1.1a:</b> Define and discuss the principle concepts of recombinant DNA technology.<br><b>11.1.1b:</b> Describe the basic tools and techniques utilized in recombinant DNA technology.                                                                                                | Student's master the ability and knowledge to: <ul style="list-style-type: none"> <li>Analyze and interpret the basic concepts of biotechnology</li> <li>Generally describe the concept of recombinant DNA technology and applications</li> <li>Evaluate and appreciate how industries are utilizing biotechnologies to improve production.</li> </ul> | <b>Written Test</b> |
| 2. Industrial Biotechnology        | <b>11.1.2a:</b> Identify and describe the applications of recombinant DNA technology in the medical or pharmaceutical and agricultural industries.<br><b>11.1.2b:</b> Appreciate the contribution of biotechnology in the improvements and advancement of health care and crop production. |                                                                                                                                                                                                                                                                                                                                                        |                     |

| Thema                              | Health and Medicine                                                                                                                                                             | Broad Content Standard                                                                                                                                                                                                                                                                                                                                                                                                         |                     |
|------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|
| <b>Module 11.1</b>                 | Health and Medical Biology                                                                                                                                                      | <b>11.1:</b> Identify and generally describe the key areas in medical and health sciences.                                                                                                                                                                                                                                                                                                                                     |                     |
| Topics                             | Performance Indicators<br>Students are expected to:                                                                                                                             | Module Benchmark                                                                                                                                                                                                                                                                                                                                                                                                               | Assessment Tasks    |
| 1. Recombination of DNA Technology | <b>11.1.1a:</b> Describe how a drug is being administered, absorbed, metabolized and excreted from the body.<br><b>11.1.1b:</b> Interpret how a drug works and its side effects | Students master the ability and knowledge to: <ul style="list-style-type: none"> <li>Describe the basic concepts of pharmacology</li> <li>Describe the common medical diagnostic investigative tools and briefly interpret the results</li> <li>Describe the SARS-CoV2-2019 Pandemic from the public health perspective and its impacts (health, economic, social and psychological) at a global and local context.</li> </ul> | <b>Written Test</b> |
| 2. Industrial Biotechnology        | <b>11.1.2:</b> Interpret and describe basic investigative radiological tools and laboratory tests used for medical diagnosis                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                |                     |
| 3. Public Health                   | <b>11.1.3:</b> Interpret and describe the public health aspects and impacts (Health, economic, social and psychological) of SARS-CoV2–2019 (COVID-19) Pandemic.                 |                                                                                                                                                                                                                                                                                                                                                                                                                                |                     |

**Table 1.3 Mapping Content Progressions by Modules**

| <b>Thema/Module</b>                                                      | <b>Broad Content Standard</b>                                                                                                                                                                               | <b>11.1</b> | <b>11.2</b> | <b>11.1</b> | <b>11.1</b> |
|--------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|-------------|-------------|-------------|
| <b>Biodiversity/</b><br>11.1 Agricultural Biodiversity                   | <b>11.1:</b> Recognize and appreciate the rich biodiversity within Papua New Guinea and how to industrialize our biodiversity cautiously and conservatively from outside influence.                         | ✓           |             |             |             |
| <b>Biodiversity/</b><br>11.2 Ornithology                                 | <b>11.2:</b> Understand the significance of Birds of Paradise to Papua New Guineans and glean the importance of the behaviours of birds and their implications in geopolitics and organizational strategies |             | ✓           |             |             |
| <b>Biotechnology/</b><br>11.1 Concepts and Applications of Biotechnology | <b>11.1:</b> Appreciate the concepts and applications of biotechnology in the medical or pharmaceutical and agricultural industries.                                                                        |             |             | ✓           |             |
| <b>Health and Medicine/</b><br>11.1 Health and Medical Biology           | <b>11.1:</b> Identify and generally describe the key areas in medical and health sciences.                                                                                                                  |             |             |             | ✓           |

# **THEMA BIODIVERSITY**

# INTRODUCTION

Biodiversity is short for biological diversity. It includes all organisms, species, and populations; the genetic variation among these; and all their complex assemblages of communities and ecosystems. It also refers to the interrelatedness of **genes**, **species**, **ecosystems** and **functions** and their interactions with the environment. Biodiversity is a key measure of the health of any ecosystem, and of our entire planet. Every organism in an ecosystem, or biome, relies on other organisms and the physical environment. For example, plant and animal species need each other for food, and depend on the environment for water and shelter. Biodiversity describes how much variety an ecosystem has, in terms of resources and species and also genetically within species. A more diverse ecosystem will have more resources to help it recover from famine, drought, disease or even the extinction of a species. There are several levels of biodiversity, each indicating how diverse the genes, species and resources are in a region that enable functioning in an ecosystem.

## Genetic Diversity

Genetic diversity is all the different genes contained in all individual plants, animals, fungi, and microorganisms. It describes how closely related the members of one species are in a given ecosystem. In simple terms, if all members have many similar genes, the species has low genetic diversity. Because of their small populations, endangered species may have low genetic diversity due to inbreeding. This can pose a threat to a population if it leads to inheritance of undesirable traits or makes the species more susceptible to disease. Having high genetic diversity helps species adapt to changing environments.

## Species Diversity

Species diversity is all the differences within and between populations of species, as well as between different species. Every ecosystem contains a unique collection of species, all interacting with each other. Some ecosystems may have many more species than another. In some ecosystems, one species has grown so large that it dominates the natural community. When comparing the biodiversity of ecosystems, an ecosystem that has a large number of species, but no species greatly outnumbering the rest, would be considered to have the most species diversity. A large number of species can help an ecosystem recover from ecological threats, even if some species go extinct.

## Ecosystem Diversity

Ecosystem diversity is all the different habitats, biological communities, and ecological processes, as well as variation within individual ecosystems. A region may have several ecosystems, or it may have one. Wide expanses of oceans or deserts would be examples of regions with low ecological diversity. A mountain area that has lakes, forests and grasslands would have higher biodiversity, in this sense. A region with several ecosystems may be able to provide more resources to help native species survive, especially when one ecosystem is threatened by drought or disease.

**Functional Diversity**

The way species behave, obtain food and use the natural resources of an ecosystem is known as *functional diversity*. In general a species-rich ecosystem is presumed to have high functional diversity, because there are many species with many different behaviors. Understanding an ecosystem's functional diversity can be useful to ecologists trying to conserve or restore damaged it, because knowing the behaviors and roles of species can point to gaps in a food cycle or ecological niches that are lacking species.



# MODULE 11.1: AGRICULTURAL BIODIVERSITY

3 weeks

*“Agricultural biodiversity is a broad term that includes all components of biological diversity of relevance to food and agriculture, and all components of biological diversity that constitute the agricultural ecosystems, also named agro-ecosystems: the variety and variability of animals, plants and micro-organisms, at the **genetic, species, ecosystem** and **functional** levels, which are necessary to sustain key functions of the agro-ecosystem, its structure and processes.”<sup>1</sup>*

This module covers the diversity of the agricultural crops native to Papua New Guinea (PNG) as a subset of the nation’s abundant biodiversity. The people of Papua New Guinea started the global Agricultural Civilization in the Kuk Valley in Western Highlands Province where drainage, planting, digging and staking of plants have been used to cultivate taro, banana and yam between 7,000 – 10,000 years ago. These agricultural practices have propagated throughout different societies that settled in PNG dating back to 9,000 years. As a result, PNG has the world’s largest agricultural biodiversity and therefore the world’s largest agricultural gene pool.

A small number of plant species have been brought from the Americas by Europeans and introduced to Indonesia between the 17th to 19th centuries from where they spread to New Guinea. This includes the sweet potato, cassava and tobacco. In the second part of the 19th century especially after 1870 further crops have been introduced directly by Europeans, including beans, pumpkin, corn, watermelon, papaya, mangosteen, durian, orange, lemon, coffee, lime and guava. Apart from that, Papua New Guinea is richly diverse in its agricultural biodiversity. Since then till today the agricultural biodiversity of Papua New Guinea has never been fully documented to create a genetic database for further research and innovations.

## Linking Previous Module (STEM Bridging)

Pre-requisite

- Ecology Module 6

## Broad Content Standard

**11. 1:** Recognize and appreciate the rich biodiversity within Papua New Guinea and how to industrialize our biodiversity cautiously and conservatively from outside influence.

## Content Benchmark

Student’s master the ability and knowledge to:

- Identify and describe the arms of agricultural biodiversity that contribute genetic resources for food and agriculture as well as ecosystem support services.
- Evaluate and analyze the vast genetic diversity within Papua New Guinea and by what means to conserve it as well as for purposes of food security and enhancing productivity.

---

<sup>1</sup> World Convention of Biodiversity, 2008.

## Topic 1: Agricultural Biodiversity in Papua New Guinea (PNG)

### Performance Indicators

Students are expected to:

11.1.1a: Describe the four levels of organisation of biodiversity.

11.1.1b: Describe the diverse agricultural biodiversity of Papua New Guinea

11.1.1c: Identify and describe the wild relatives of agricultural crops native to Papua New Guinea

11.1.1d: Identify and describe the limitations to agricultural development in Papua New Guinea.

### Content Expansion

#### ***Body of Knowledge for their understanding***

- The four levels of biodiversity
  - Genetic Diversity
  - Species Diversity
  - Ecosystem Diversity
  - Functional Diversity
- Papua New Guinea's Biodiversity
  - Flora and Fauna
- Agricultural Biodiversity in Papua New Guinea
  - Start of Agricultural Civilization in PNG
- Agricultural Crops that are native to PNG:
  - Root/Tuber crops (Eg. Taro)
  - Green Leafy Vegetables (Eg. Aibika, etc...)
  - Banana
  - Fruits (Eg. Mondon, etc...)
  - Nuts (eg. Betelnut, etc...)
  - Beans (eg winged beans )
- Wild relatives of agricultural crops in PNG
  - Wild Taro
  - Wild Rice
- The limitations of Agricultural Practice in Papua New Guinea
  - Why agricultural practice never advanced in PNG?! (Beast of burdens)
  - Volcanic Eruption of Mt. Hagen and redirection of Waghi River into the Kuk Valley after the eruption resulting in migration of settlers up into the mountains.

#### ***Set of Skills for their proficiency levels***

- Describe the four levels of biodiversity and how they inter-relate or coexist.
- Describe the rich biodiversity of Papua New Guinea
- Describe Papua New Guinea's agricultural biodiversity and
- Identify wild relatives of agricultural crops native to PNG
- Identify the limitations to agricultural development in PNG

***Trait of Attitudes and Values to govern their lives***

- Value the rich biodiversity and more to agricultural biodiversity of our Country.
- Appreciate the beginning of agriculture in Papua New Guinea 10,000 years ago and the reason we were stuck, when the rest of the world advanced in agriculture.
- Appreciate the relevance of taxonomy in agriculture.

## Topic 2: Genetic Diversity

### Performance Indicators

Students are expected to:

- 11.1.2a: Identify and describe the genetic diversity of native agricultural crops in Papua New Guinea compared to the world and how to conserve native species.
- 11.1.2b: Recognize the importance of genetic sequencing of native agricultural crops in PNG and their applications to address climate resilience and food security.

### Content Expansion

#### ***Body of Knowledge for their understanding***

- The reason PNG remained genetically diverse
- The Genetic Diversity of Native Agricultural Crops in PNG
  - Taro
  - Yam
  - Sweet Potato
  - Sugar cane
  - Banana
- The importance of Genetic Sequencing of Agricultural Crops

#### ***Set of Skills for their proficiency levels***

- Identify a certain native agricultural crops to demonstrate the genetic diversity within Papua New Guinea.
- Understand the importance of genetic sequencing of native agricultural crops in PNG and their applications to address climate resilience and food security.
- Document them according to their taxonomy.
- Discuss and present their results/findings

#### ***Trait of Attitudes and Values to govern their lives***

- Appreciate the genetic diversity of native agricultural crops in Papua New Guinea
- Propose for potential genetic manipulation for industrial purposes.

### Application of Knowledge and Skills (Career Focus)

- Biologists
- Biochemists
- Ethnobotanists
- Chemists
- Pharmacists
- Genetic engineers

## MODULE 11.2: ORNITHOLOGY

3 weeks

Ornithology is a branch of zoology that deals with the study of birds. Studies of birds can be useful to develop key concepts in biology including behaviour, ecology, distributions of species, ecological niches, and conservation. A wide range of tools and techniques are used in ornithology, both inside the laboratory and out in the field, and innovations are constantly made.

In this module, for the first time in the history of Papua New Guinea, students will study the Birds of Paradise because of their national significance (Topic 1). The second topic covers inspirations from bird behaviour that are of geopolitical significance and corporate predatory behaviour.

### Linking Previous Module (STEM Bridging)

Pre-requisite: NIL

### Broad Content Standard

**11.2:** Understand the significance of Birds of Paradise to Papua New Guineans and glean the importance of the behaviours of birds and their implications in geopolitics and organizational strategies.

### Content Benchmark

Students are expected to master the ability and knowledge to:

- Describe the ornithology of the Birds of Paradise species and their significance in Papua New Guinea.
- Discuss and analyze the geopolitical significance of inspirations from bird behaviors that are unique to human or organizational strategies.

## Topic 1: Birds of Paradise

### Performance Indicators

Students are expected to:

11.2.1: Analyze the importance of the Ornithology of the Birds of Paradise as our National Emblem, the species diversity and their natural character or behaviour.

### Content Expansion

#### ***Body of Knowledge for their understanding***

- Papua New Guinea's Biodiversity of Birds (Ornithology)
  - Cassowaries
  - Pigeons and Parrots
  - Passerines
  - Bowerbirds
  - Hooded Pitohui
  - Harpy Eagle
- Birds of Paradise
  - Taxonomy
  - Description
  - Habitat and distribution
  - Behaviour and ecology

#### ***Set of Skills for their proficiency levels***

- Describe PNG's biodiversity of birds
- Identify the taxa of the birds of paradise
- Describe the importance of the Ornithology of the Birds of Paradise as our National Emblem, the species diversity and their natural character or behaviour.
- Discuss case studies of Bird of Paradise

#### ***Trait of Attitudes and Values to govern their lives***

- Examine the importance of the ornithology of birds
- Value different species of birds of paradise
- Appreciate the birds of paradise as national emblem
- Appreciate the behaviour of the birds of paradise and their behaviour embedded in society.

## Topic 2: Inspirations from Bird Behavior

### Performance Indicators

Students are expected to:

- 11.2.2a: Analyze the nesting and predatory behaviour of birds that inspire geopolitical strategies utilized by organizations.
- 11.2.2b: Appreciate the inspirations from bird behaviour that has significant applications in diverse settings.
- 11.2.2c: Give examples: Mendondi and Tenduage Strategy as applied in Organizational and Corporate Strategies.

### Content Expansion

#### ***Body of Knowledge for their understanding***

- Brood Parasitology (Mendondi-Tenduage Strategy)
  - Adaptations for parasitism
  - Nest-site hypothesis
  - Parental-care parasitism
- **Geopolitical Significance**
  - Nesting behavior in institutions
    - *Case study on NARI*
  - Predatory behavior in corporate boardroom
    - *Case study on Bmobile*

#### ***Set of Skills for their proficiency levels***

- Discuss case studies of inspirations from bird behavior
- Relate and present findings of bird behavior to organizational strategies.

#### ***Trait of Attitudes and Values to govern their lives***

- Appreciate the inspirations from bird behaviour that has significance to geopolitics and organizational strategies.
- Appreciate birds behaviour as one of the keys to understanding cultural heritage and why people behave and act.

## Application of Knowledge and Skills (Career Focus)

- Ornithologists
- Biologists
- Biochemists
- Ethnobotanists
- Chemists
- Pharmacists
- Genetic engineers

# **THEMA BIOTECHNOLOGY**



# INTRODUCTION

Biotechnology combines biology with technology to engineer, modify or upgrade the parts of biological system for industrial and human welfare. It combines various technologies derived from biology, mathematics, physics, chemistry and engineering to modify the biological systems. Its application ranges from agriculture to industry, medicine, nutrition, environmental conservation, cell biology, and so on. Biotechnology is now one of the fastest growing fields of sciences.

## MODULE 11.1: CONCEPTS AND APPLICATIONS OF BIOTECHNOLOGY

*3 weeks*

This module introduces the fundamental concepts of biotechnology and its applications in the medical or pharmaceutical, agricultural and industrial fields. There are presently more than 250 health care products derived through biotechnology. There are also vaccines available for diseases that were previously untreatable. A topical example is the coronavirus vaccine developed by the major pharmaceutical companies for SARS-CoV2-2019.

There are also applications in agricultural biotechnology to increase crop yields, prevent damage from pests and diseases, and reduce environmental impacts by farming. In addition, there are several biorefineries built around the world to produce biofuels from biomass to mitigate greenhouse gases and address Global Warming and Climate Change.

### Linking Previous Module (STEM Bridging)

Pre-requisite

- 11.1 Introduction to Biology
- 11.2 Cells
- 11.5 Genetics

### Broad Content Standard

**11.1:** Appreciate the concepts and applications of biotechnology in the medical or pharmaceutical and agricultural industries.

### Content Benchmark

Student's ability to master knowledge and skills of:

- The basic concepts of biotechnology
- The concept of recombinant DNA technology and its applications
- Utilizing biotechnology to improve production with enzymes and microbes as well as saving energy by converting the biomass into biofuels by industries.

## Topic 1: Recombinant DNA Technology

### Performance Indicators

Students are expected to:

11.1.1a: Define and discuss the principle concepts of recombinant DNA technology.

11.1.1b: Describe the basic tools and techniques utilized in recombinant DNA technology.

### Content Expansion

#### ***Body of Knowledge for their understanding***

- Recombinant DNA technology
  - DNA cloning
  - Restriction Sites and Restriction Enzymes
  - DNA Amplification - Polymerase Chain Reaction (PCR)
  - Recombinant Expression - Production of Proteins

#### ***Set of Skills for their proficiency levels***

- Identify and describe the basic principles or concepts in biotechnology.
- Describe the recombinant DNA technology and how it works to produce genetically modified organisms

#### ***Trait of Attitudes and Values to govern their lives***

- Appreciate the advent of biotechnology with recombinant DNA technology.
- Appreciate the vast applications of recombinant DNA technology in health care and agriculture.

## Topic 2: Industrial Biotechnology

### Performance Indicators

Students are expected to:

- 11.1.2a: Identify and describe the applications of recombinant DNA technology in the medical or pharmaceutical and agricultural industries.
- 11.1.2b: Appreciate the contribution of biotechnology in the improvements and advancement of health care and crop production.

### Content Expansion

#### ***Body of Knowledge for their understanding***

- Genetically Engineered or Modified Organisms
- Transgenic Animals
- Improving Production and Disease Resistance
- Gene Therapy
- Diagnostics
- Genomics
- Protein Manufacture

#### ***Set of Skills for their proficiency levels***

- Discuss aspects of recombinant DNA technology
- Discuss how recombinant DNA technology works in manipulating the DNA and genes of interest.
- Describe the applications of biotechnology in the medical or pharmaceutical and agricultural industries.

#### ***Trait of Attitudes and Values to govern their lives***

- Appreciate the benefits of biotechnology in terms of improved production, food security and enhanced living standard.
- Appreciate the use of recombinant DNA technology in industries such as the medical or health (pharmaceutical), agriculture and others.
- Value the importance of biotechnology in pharmaceutical and agricultural industries

### Application of Knowledge and Skills (Career Focus)

- Genetic engineers
- Biologists
- Biochemists
- Biotechnologists
- Biological Engineers
- Chemical Engineers
- Chemists

# **THEMA HEALTH AND MEDICINE**

# INTRODUCTION

Medicine is the art, science and practice of caring for a patient and managing the diagnosis, prognosis, prevention, treatment or palliative care of their injury or diseases. Medicine encompasses health care practices evolved to maintain and restore health by the prevention and treatment of illness. The practice of medicine has significantly changed with the advent of molecular biology with its enormous implications in biological sciences (sequencing of human genome), sophisticated new imaging techniques and advances in bioinformatics and information technology. That has sparked an explosion of scientific information that fundamentally influenced and changed the way in which patient care and management has been practiced.

The knowledge gleaned from the science of medicine has already improved and undoubtedly will further improve our understanding of complex disease processes and provide new approaches to disease treatment and prevention. Though the practice of medicine requires a scientific basis, the combination of medical knowledge, intuition, experience and judgment defines the art of medicine.

Furthermore, the recent emergence of SARS-CoV2–2019 (COVID-19) pandemic has shut down the world and caused severe economic downturn globally; it has disrupted the global supply chain. It has exposed the deficiency within the Papua New Guinean health delivery services and their political decision-making processes. This is not the first global outbreak of novel viral diseases. There have been similar outbreaks in the recent past, such as Middle-East Respiratory Syndrome (MERS), Severe Acute Respiratory Syndrome (SARS), Avian Influenza (H1N1 and its variants), African Swine Flu, and so on. It is therefore of paramount importance to educate and prepare the next generation of medical doctors and scientists with sound knowledge in medicine to combat emergence of future pandemics.

## MODULE 11.1: MEDICINE AND HEALTH SCIENCES

*3 weeks*

In this module necessary concepts of study in the field of medicine and health sciences would be covered. This includes the notions of Pharmacology in Topic 1 which would cover basic concepts of how drugs work and interestingly how drugs can be simply designed with the advent of technology. In Topic 2, the medical diagnostic investigative tools such as the radiological studies and laboratory tests to confirm or exclude varying disease processes or injuries would be covered. Moreover, with the trending issue of Coronavirus (COVID-19) Pandemic, understanding its impacts and implications in the health, economic, social and most importantly the geopolitical aspects of it would assist our government to make better informed decisions. It is this part of public health and policy making that creates the forefront of preventative medicine and health care practice that is paramount to governing our health sector and is therefore introduced here in Topic 3.

### Linking Previous Module (STEM Bridging)

Prerequisite:

- Module 11.1 Introduction to Biology
- Module 11.2 Cells
- Module 11.3 Metabolism
- Module 11.4 Anatomy and Physiology

### Broad Content Standard

**11.1:** Identify and generally describe the key areas in medical and health sciences.

### Content Benchmark

Students master the ability and knowledge to:

- Describe the basic concepts of pharmacology
- Describe the common medical diagnostic investigative tools and briefly interpret the results
- Describe the SARS-CoV2-2019 Pandemic from the public health perspective and its impacts (health, economic, social and psychological) at a global and local context.

## Topic 1: Pharmacology

### Performance Indicators

Students are expected to:

11.1.1a: Describe how a drug is being administered, absorbed, metabolized and excreted from the body.

11.1.1b: Interpret how a drug works and its side effects

### Content Expansion

#### ***Body of Knowledge for their understanding***

- How a Medical Drug Works?
  - Route of Drug Administration
  - Drug Absorption
  - Drug Bioavailability
  - Drug distribution
  - Drug biotransformation
  - Drug excretion
  - Drug mechanism of action
  - Drug side effects

#### ***Set of Skills for their proficiency levels***

- Describe how a medical drug works from the point of administration to its effects.

#### ***Trait of Attitudes and Values to govern their lives***

- Value and be aware of how medical drugs work
- Appreciate how drugs metabolized in the body and excreted
- Recognize the desired and undesired effects of drugs

## Topic 2: Medical Diagnostic Investigations

### Performance Indicators

Students are expected to:

11.1.2: Interpret and describe basic investigative radiological tools and laboratory tests used for medical diagnosis

### Content Expansion

#### ***Body of Knowledge for their understanding***

- Radiological Investigations
- Principles behind the Imaging Modalities and Interpretation
  - Ultrasound
  - X-ray Imaging
- Laboratory Investigations
- Principles behind the laboratory tests and Interpretation
  - Cytogenetics And Molecular Genetics
  - Genetic Testing
- Diagnostic testing includes:
  - Cell-free fetal DNA (cffDNA) testing
  - Newborn screening
  - Diagnostic testing
  - Carrier testing
  - Preimplantation genetic diagnosis
  - Prenatal diagnosis
  - Predictive and presymptomatic testing
  - Pharmacogenomics
- Non-diagnostic testing includes:
  - Forensic testing
  - Paternity testing
  - Genealogical DNA test
  - Research testing

#### ***Set of Skills for their proficiency levels***

- Discuss and describe the importance of ultrasound and x-ray imaging and principles behind tests and interpretation
- Discuss and describe the importance of genetic testing and its principles

#### ***Trait of Attitudes and Values to govern their lives***

- Appreciate the different medical diagnostic investigative tools
- Value the knowledge gained to generally interpret the results from the radiological and laboratory investigations.



## Topic 3: Public Health

### Performance Indicators

Students are expected to:

11.1.3: Interpret and describe the public health aspects and impacts (Health, economic, social and psychological) of SARS-CoV2–2019 (COVID-19) Pandemic.

### Content Expansion

#### *Body of Knowledge for their understanding*

- SARS-CoV2–2019 (COVID-19) Pandemic
  - Health Impacts
  - Economic Impacts
  - Social and Psychological Impacts
  - Political Decision-Making

#### *Set of Skills for their proficiency levels*

- Describe the impacts of the SARS-CoV2-2019 pandemic in the health, economic, social and psychological, and the political decision making process.

#### *Trait of Attitudes and Values to govern their lives*

- Appreciate public health as a key area in the first line of defence of the health of this country and in guiding policy making.
- Appreciate the impacts of SARS-CoV2-2019 Pandemic globally from the developed countries to the developing countries and its implications.
- Be aware of the implications of such pandemics as COVID-19 and how to manage future pandemics.

### Application of Knowledge and Skills (Career Focus)

- Medical Doctors
- Pharmacists
- Laboratory Scientists
- Radiologists
- Nurses
- Public Health Administrators
- Biologists
- Biochemists
- Chemists
- Genetic engineers

## REFERENCES

1. Bolger J. Sciencing: four types of biodiversity. Leaf Group Ltd. March 09, 2018. (Accessed online 18-02-21: <https://sciencing.com/difference-between-community-ecosystem-5776576.html>)
2. Convention on biological diversity. What is Agricultural Biodiversity? UN environment programme. (Accessed online 15-02-21: <https://www.cbd.int/agro>)